

UNIVERSITY OF WATERLOO, ON, Canada

WMO: 713685

Lat: 43.4735N

Lon: 80.5572W

Elev: 1096

StdP: 14.12

Time Zone: -5.00 (NAE)

Period: 98-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-5.9	0.8	-10.0	3.3	-4.5	-3.0	4.9	1.0	15.6	26.8	14.1	27.9	1.0	320	0.458

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.8	86.9	73.9	84.2	72.5	81.5	70.9	77.2	82.9	75.0	81.0	73.2	79.0	4.2	260

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
75.5	139.3	81.1	73.2	128.7	78.4	71.4	120.6	76.6	41.5	82.9	39.3	81.1	37.6	79.0	88.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
(a)	(b)	(c)	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
13.5	11.6	9.9	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
			-13.8	91.1	6.9	3.8	-18.8	93.8	-22.8	96.0	-26.6	98.1	-31.6	100.8
			-15.4	79.9	7.8	3.5	-21.1	82.4	-25.6	84.4	-30.0	86.3	-35.7	88.9
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	45.9	20.7	21.8	31.0	43.3	55.5	64.9	69.2	67.7	61.3	48.9	37.9	26.9
	DBStd	19.37	11.62	10.82	11.10	8.95	8.52	7.02	5.60	5.38	7.34	8.34	8.30	9.76
	HDD50	3787	907	791	599	237	37	1	0	0	4	124	370	717
	HDD65	7403	1372	1211	1054	652	317	86	22	34	158	502	813	1182
	CDD50	2286	0	0	10	36	209	448	595	548	343	90	7	0
	CDD65	427	0	0	0	1	23	84	152	117	46	4	0	0
	CDH74	3631	0	0	3	14	231	752	1320	887	395	29	0	0
	CDH80	888	0	0	0	0	44	198	357	195	91	3	0	0
Wind	WSAvg	3.7	4.5	4.5	4.7	4.8	4.0	3.0	2.5	2.4	2.5	3.1	3.8	4.1
Precipitation	PrecAvg	38.1	3.4	2.6	2.6	3.1	3.6	3.2	3.9	2.7	3.3	3.1	3.4	3.0
	PrecMax	46.8	5.1	4.2	4.4	5.6	6.4	6.9	6.9	5.6	9.0	6.1	6.4	5.5
	PrecMin	26.7	1.8	1.2	1.3	1.3	1.5	1.7	0.6	0.6	1.0	0.9	1.5	1.6
	PrecStd	5.2	1.1	1.0	1.0	1.1	1.2	1.2	1.6	1.1	1.9	1.4	1.3	0.9
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	51.6	53.2	70.1	76.3	84.9	88.2	90.9	89.2	87.1	77.7	63.7	55.1
		MCWB	51.1	49.6	62.9	63.5	70.7	74.3	77.8	74.8	73.4	67.4	57.2	53.1
	2%	DB	45.4	45.6	59.8	69.4	80.1	85.3	87.2	84.7	82.5	71.3	58.5	48.6
		MCWB	44.3	43.4	53.3	59.0	67.9	72.1	75.0	73.1	71.1	63.7	54.6	47.0
	5%	DB	39.7	40.2	53.6	64.5	76.1	82.1	83.9	81.7	78.6	67.0	55.1	43.6
		MCWB	38.5	37.7	49.2	55.1	64.4	70.8	72.6	71.9	68.9	61.4	51.8	42.1
	10%	DB	35.7	35.8	47.5	59.4	71.3	78.4	81.0	79.0	74.7	62.6	51.2	39.2
		MCWB	34.8	34.3	43.5	51.3	61.1	68.7	70.8	70.3	67.0	58.4	48.5	38.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	51.2	51.6	63.6	66.5	74.7	77.7	81.5	78.9	77.2	70.5	58.9	53.3
		MCDB	51.4	52.5	69.2	71.3	80.8	83.9	88.6	82.8	83.4	75.1	61.4	54.0
	2%	WB	44.7	44.2	55.0	61.5	70.1	74.7	78.0	75.9	73.7	65.8	55.6	47.5
		MCDB	45.1	46.0	58.9	67.4	76.5	81.7	83.6	81.8	79.7	69.2	58.0	48.5
	5%	WB	38.7	38.0	49.7	57.3	66.7	72.8	75.4	73.7	71.0	62.2	52.1	42.3
		MCDB	39.6	40.0	53.3	62.3	73.1	79.4	80.6	79.3	75.7	66.1	54.5	43.3
	10%	WB	34.8	34.5	43.7	52.4	63.2	70.7	73.1	71.9	68.5	58.9	48.7	38.1
		MCDB	35.6	35.9	47.1	58.1	69.7	76.5	78.6	76.7	73.1	62.3	51.0	39.3
Mean Daily Temperature Range	5% DB	MDBR	14.2	15.9	16.5	19.0	19.8	19.1	19.8	19.2	20.1	16.7	13.6	11.6
		MCDBR	15.9	19.1	25.0	28.1	24.4	22.7	22.8	21.8	24.7	23.1	20.6	15.2
		MCWBR	15.2	16.9	18.6	18.7	14.2	12.2	12.1	11.8	14.0	15.6	16.6	14.0
	5% WB	MCDBR	15.3	18.1	23.5	23.9	21.8	19.9	19.6	18.5	20.1	20.8	18.7	15.1
		MCWBR	14.9	16.5	18.7	19.0	15.0	12.3	12.0	11.7	13.1	15.1	16.2	14.2
Clear-Sky Solar Irradiance	taub	0.286	0.331	0.325	0.360	0.386	0.417	0.409	0.397	0.368	0.351	0.305	0.284	
	taud	2.366	2.182	2.389	2.335	2.306	2.235	2.278	2.297	2.374	2.409	2.502	2.437	
	Ebn at Noon	268	269	291	287	281	272	272	272	272	262	261	259	
	Edh at Noon	25	36	33	38	40	43	41	39	33	29	22	21	
All-Sky Solar Radiation	RadAvg	443	714	1116	1412	1708	1891	1878	1638	1313	786	482	318	
	RadStd	59	91	116	160	163	124	126	83	126	75	67	42	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (52 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page